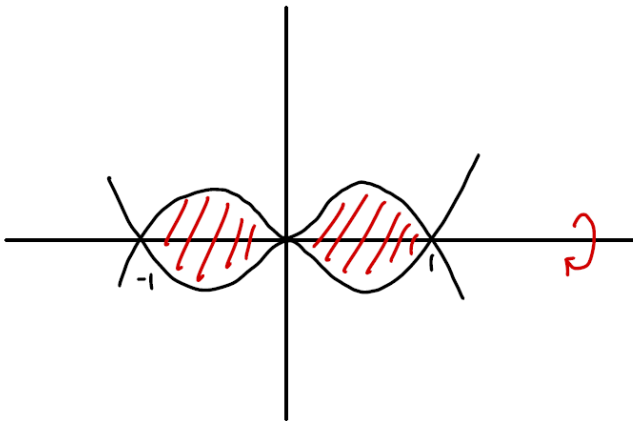


1



$$V = 2 \times \pi \int_0^1 (2 - x^2) dx = \frac{4}{15} \pi //$$

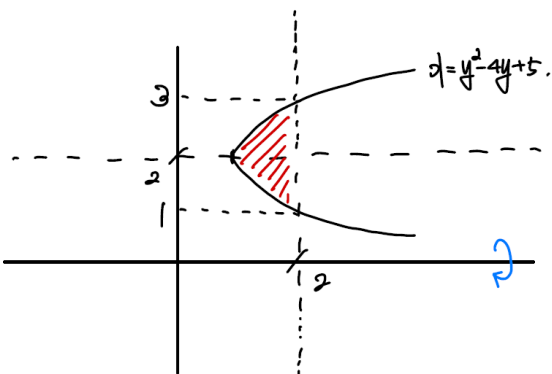
2

$$\sin^{-1} x \leq t \Rightarrow dx = \sec^2 t dt.$$

$$\int_0^{\frac{\pi}{3}} \sin t \cdot \sec^2 t dt = \int_0^{\frac{\pi}{3}} \sec t \cdot \sin t dt.$$

$$= \left[\sec t \right]_0^{\frac{\pi}{3}} = 2 - 1 = 1 //$$

3



$$V = 2\pi \cdot \int_1^3 y (2 - (y^2 - 4y + 5)) dy = \frac{16}{3} \pi //$$

4

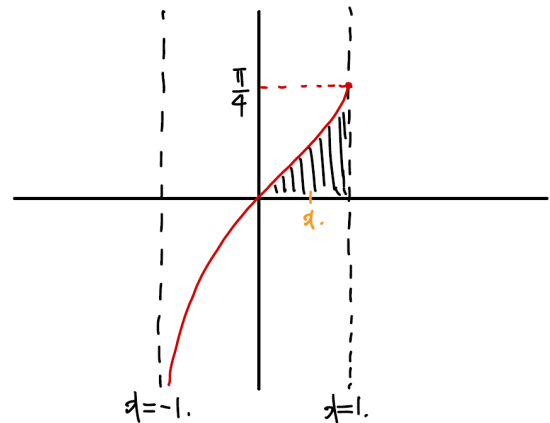
$$f'(x) = \frac{1}{\sqrt{4+x^2}} = \left\{ \sinh^{-1}\left(\frac{x}{2}\right) \right\}'$$

$$\therefore \left\{ \sinh^{-1}\left(\frac{x}{2}\right) \right\}' = \frac{1}{\sqrt{4+\frac{x^2}{4}}} \cdot \frac{1}{2} = \frac{1}{\sqrt{4+x^2}}$$

$$\therefore \sinh^{-1}\left(\frac{x}{2}\right) \Big|_{x=2} = \sinh^{-1}(1).$$

$$\ast \sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1}) \Big|_{x=1} = \ln(1 + \sqrt{2}) //$$

5



$$V = 2\pi \int_0^1 (x+1) \cdot \frac{\sin^{-1} x}{x+1} dx = 2\pi \int_0^1 \sin^{-1} x dx = \pi^2 - 2\pi //$$

$$\int_0^{\frac{\pi}{2}} \sin x dx + \int_0^1 \sin^{-1} x dx = \frac{\pi}{2}$$

$$\therefore \int_0^1 \sin^{-1} x dx = \frac{\pi}{2} - 1.$$

6

$$36 \int_1^{\sqrt{3}} \frac{x^2 - x + 6}{x(x^2 + 3)} dx$$

$$\frac{2}{x} - \frac{x}{x^2 + 3} - \frac{1}{x^2 + 3}$$

$$= 36 \left[2 \ln|x| - \frac{1}{2} \ln|x^2 + 3| - \frac{1}{\sqrt{3}} \ln \left| \frac{x}{\sqrt{3}} \right| \right]_1^{\sqrt{3}}$$

$$= 18 \ln 6 - \sqrt{3} \pi \quad \therefore a+b+c+d = 26$$

7

$$\dots \int_0^1 \frac{1}{x^p} dx + \int_1^\infty \frac{1}{x^p} dx.$$

(I₁) (I₂)

I₁: $0 < p < 1$: 수렴. $p \geq 1$: 발산.

I₂: $p > 1$: 수렴. $0 < p \leq 1$: 발산.

$\therefore$ 5번!

8

$$\lim_{n \rightarrow \infty} \frac{1}{2\sqrt{n}} \sum_{k=1}^n \frac{1}{\sqrt{n+k}} = \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{k=1}^n \frac{1}{\sqrt{1+\frac{k}{n}}}$$

$$\xrightarrow{1+\frac{k}{n} \triangleq x} \int_1^2 \frac{1}{x} \cdot \frac{1}{\sqrt{x}} dx = \sqrt{2} - 1$$

9

$$\mathcal{J} = 2\pi \int_1^4 x \sqrt{\left(\frac{x}{2} + \frac{1}{2x}\right)^2} dx = 2\pi \int_1^4 x \left(\frac{x}{2} + \frac{1}{2x}\right) dx$$

$$= \pi \int_1^4 (x^2 + 1) dx = \frac{24\pi}{3}$$

10

$$\ln x \triangleq u \Rightarrow \frac{1}{x} dx = du.$$

$$\int_0^2 e^u (1+u)^2 du.$$

$$= \left[(1+u)^2 e^u \right]_0^2 - \int_0^2 2(1+u)e^u du$$

$$= 9e^2 - 1 - 2 \left\{ \left[(1+u)e^u \right]_0^2 - \int_0^2 e^u du \right\}$$

$$= 5e^2 - 1$$

11

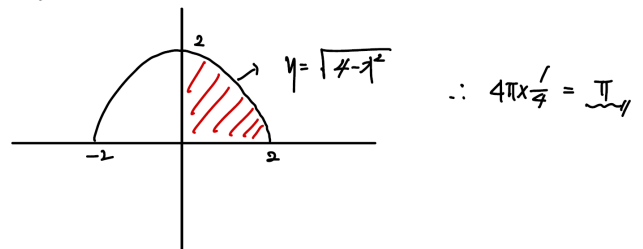
$$\frac{2k}{\pi} \triangleq a \rightsquigarrow \int_0^2 \sqrt{4-x^2} dx$$

sol I)

$$x \triangleq 2 \sin \theta \Rightarrow dx = 2 \cos \theta d\theta.$$

$$\therefore \int_0^{\frac{\pi}{2}} 4 \cos^2 \theta d\theta = \cancel{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \pi.$$

sol II)



12

$$d \equiv 25 \text{ m} \Rightarrow dx = 25 \cos \theta d\theta.$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin^2 \theta \cdot 3 \cos \theta} \cdot \cancel{3 \cos \theta} d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin^2 \theta} \cdot \cancel{\cos \theta} d\theta$$

$$= \left[-\frac{1}{9} \sin \theta \right]_{\sin^{-1} \frac{1}{3}}^{\frac{\pi}{2}} = \frac{1}{9} \sin \left(\sin^{-1} \frac{1}{3} \right).$$

$$\sin x = \frac{1}{3} \quad \begin{array}{c} 3 \\ \diagup \\ 1 \\ \diagdown \\ 2\sqrt{2} \end{array} \quad \therefore \sec x = 2\sqrt{2} \quad \therefore \frac{2}{9}\sqrt{2}$$

13

$$f' > 0 \Rightarrow f: \text{역함수 존재.}$$

$$\int_1^4 f(x) dx + \int_{f(u)}^{f(4)} f^{-1}(x) dx = 4f(4) - f(u).$$

$$= \underline{\underline{21}}$$

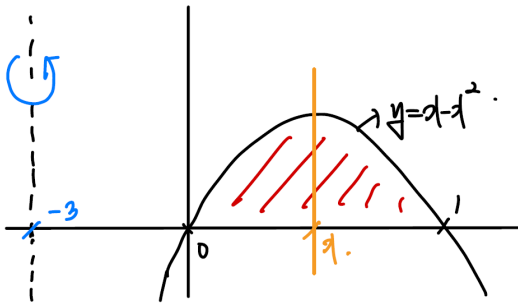
14

$$\left. \begin{aligned} \int_0^1 \frac{dx}{\sqrt{x}} &= \left[2\sqrt{x} \right]_0^1 - \int_0^1 \frac{2}{\sqrt{x}} dx = -4 \\ \int_e^\infty \frac{1}{x(\ln x)^2} dx &= \left[-\frac{1}{\ln x} \right]_e^\infty = 1 \end{aligned} \right) \Rightarrow \underline{\underline{-3}}$$

15

$$\begin{aligned} \int_0^\infty [1]e^{-x}dx &= \int_0^1 0e^{-x}dx + \int_1^2 1e^{-x}dx + \int_2^3 2e^{-x}dx + \dots \\ &= (-e^{-2} + e^{-1}) + 2(-e^{-3} + e^{-2}) + 3(-e^{-4} + e^{-3}) + \dots \\ &= e^{-1} + e^{-2} + e^{-3} + \dots = \frac{e^{-1}}{1 - e^{-1}} = \frac{1}{e-1} \end{aligned}$$

19



$$S = 2\pi \int_0^1 (x+3)(x-x^2) dx = \frac{7}{2}\pi //$$

20

$$f(g(x)) = 1 + f(g(x))' = 1 + x'.$$

$$\Rightarrow f(g(x))' = f'(g(x)) \cdot g'(x) = 1 \quad \therefore g'(x) = \frac{1}{f'(g(x))} = \frac{1}{1+x^2}$$

$$\therefore g(x) = \frac{1}{x} //$$

21

$$\int_0^{\frac{\pi}{4}} \frac{1 - 4 \tan x}{4 + \tan x} dx.$$

$$= \int_0^{\frac{\pi}{4}} \frac{\cos x - 4 \sin x}{4 \cos x + \sin x} dx. \quad \text{이분!}$$

$$= \left[\ln |\sin x + 4 \cos x| \right]_0^{\frac{\pi}{4}}$$

$$= \ln \left(\frac{\sqrt{2}}{2} + 2\sqrt{2} \right) - \ln 4.$$

$$= \ln \left(\frac{5\sqrt{2}}{2} \right) //$$

22

$$\rho \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \ln \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right).$$

$$(1 + \frac{1}{n})(1 + \frac{2}{n}) + \cdots + (1 + \frac{n}{n}).$$

$$= \rho \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right) = \rho \int_1^2 \ln x dx.$$

$$1 + \frac{k}{n} \triangleq x.$$

$$= \rho [x \ln x - x]_1^2 = \rho^{2 \ln 2 - 1} = \frac{4}{e} //$$

23

$$\sin^{-1} x \triangleq \alpha \Rightarrow x = \sin \alpha \quad \left(-\frac{\pi}{2} < \alpha < \frac{\pi}{2} \right).$$

$$0 < x < 1 \Rightarrow \left(0 < \alpha < \frac{\pi}{2} \right).$$



$$\cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \triangleq \beta \Rightarrow \frac{1}{\sqrt{1+x^2}} = \cos \beta \quad (0 \leq \beta \leq \pi).$$



$$0 < x < 1 \Rightarrow \left(0 \leq \beta \leq \frac{\pi}{2} \right).$$

$$\therefore \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta = \frac{x}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \cdot \frac{x}{\sqrt{1+x^2}} = \frac{2x}{1+x^2}$$

$$\Rightarrow \int_0^1 \frac{2x}{1+x^2} dx = \left[\ln |1+x^2| \right]_0^1 = \ln 2 //$$

24

$$\pi \int_0^{\pi} \sin^2 x dx \leq \pi \int_0^{\pi} e^{\sin x} dx \leq \pi \int_0^{\pi} e^x dx$$

(가) (나) (다).

($\because \sin x \leq e^{\sin x} \leq e^x, 0 \leq x \leq \pi$).

25

sol I).

$$\frac{2(x^2+1)}{x^4+2x^2-1} = \frac{2(x^2+1)}{(x^2+1)^2 - x^2} = \frac{2x^2+2}{(x^2-x+1)(x^2+x+1)}$$

$$\rightsquigarrow \frac{0}{0} \text{ 형태 } \frac{A}{B} + \frac{C}{D}$$

$$\rightsquigarrow \frac{1}{(x-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} + \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}$$

$$\int \frac{1}{(x-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} + \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dx$$

$$= \frac{2}{\sqrt{3}} \ln \left(\frac{2}{\sqrt{3}} \cdot (x-\frac{1}{2}) \right) + \frac{2}{\sqrt{3}} \ln \left(\frac{2}{\sqrt{3}} \cdot (x+\frac{1}{2}) \right) + C.$$

* $\ln^+ x + \ln^+ y = \ln^+ \left(\frac{x+y}{1-xy} \right)$. 이용!

$$\rightsquigarrow \frac{2}{\sqrt{3}} \left(\ln^+ \left(\frac{2}{\sqrt{3}} \cdot (x-\frac{1}{2}) \right) + \ln^+ \left(\frac{2}{\sqrt{3}} \cdot (x+\frac{1}{2}) \right) \right)$$

$$\ln^+ \left(\frac{\frac{4}{\sqrt{3}}x}{1 - (\frac{4}{3}x^2 - \frac{1}{3})} \right) = \ln^+ \left(\frac{\frac{4}{\sqrt{3}}x}{\frac{4}{3} - \frac{4}{3}x^2} \right)$$

$$= \ln^+ \left(\frac{\sqrt{3}x}{1-x^2} \right).$$

$$\therefore \alpha = \sqrt{3}, \beta = -1.$$

sol II)

$$\frac{2(x^2+1)}{x^4+2x^2-1} = \frac{2x^2+2}{x^4-2x^2+3x^2} \Rightarrow \int \frac{2(x^2+1)}{(x^2-1)^2 + (\sqrt{3}x)^2} dx$$

$$\Rightarrow \int \frac{\frac{2(x^2+1)}{(x^2-1)^2}}{1 + (\frac{\sqrt{3}x}{x^2-1})^2} dx \dots \rightarrow \frac{\sqrt{3}x}{x^2-1} = t \Rightarrow -\sqrt{3} \cdot \frac{x^2+1}{(x^2-1)^2} dx = dt$$

$$\Rightarrow \int \frac{2}{4t^2} \left(-\frac{1}{\sqrt{3}} \right) dt = -\frac{2}{\sqrt{3}} \ln |t| + C = \frac{2}{\sqrt{3}} \ln \left(\frac{\sqrt{3}x}{1-x^2} \right) + C$$

26

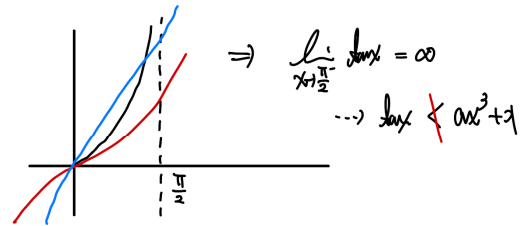
$$g(x) = x \cdot \int_0^1 f(t) dt - \int_0^1 t f(t+x) dt$$

$$g(x) = \int_0^1 f(t) dt + x f'(x) - x f'(x)$$

$$= \tan x - ax^3 - x + x^0 (\because g(0)=0).$$

$\neq 0$. $\begin{matrix} >0 \\ <0 \end{matrix}$

$$i) \tan x < ax^3 + x = x(ax^2+1), x=0 (\because a>0).$$



$$ii) \tan x > ax^3 + x.$$

 $\frac{d}{dx}$

$$\sec^2 x = 3ax^2 + 1 \rightsquigarrow x=0.$$

$$\therefore \lim_{x \rightarrow 0} \frac{Mac}{x} = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$$

$$(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots) - (ax^3 + x) > 0.$$

$$\rightsquigarrow (\frac{1}{3} - a)x^3 + \frac{2}{15}x^5 + \dots > 0 \quad \therefore a_{max} = \frac{1}{3}.$$

27

$$\begin{aligned}
 I_{n+2} &= \int_0^{\pi} \frac{\sin(n+2)x}{\sin x} dx. \\
 &= \int_0^{\pi} \frac{\sin nx \cdot \cos 2x + \cos nx \cdot \sin 2x}{\sin x} dx. \\
 &= \int_0^{\pi} \frac{\sin nx}{\sin x} - 2\sin nx \cdot \cos x + 2\cos nx \cdot \cos x dx. \\
 &= I_n + 2 \int_0^{\pi} (\cos nx \cos x - \sin nx \sin x) dx = I_n + 2 \int_0^{\pi} \cos(n+1)x dx \\
 &= I_n + 2 \left[\frac{1}{n+1} \sin(n+1)x \right]_0^{\pi} = I_n \\
 \therefore I_{n+2} &= I_n. \\
 \therefore I_1 &= \int_0^{\pi} \frac{\sin x}{\sin x} dx = \pi. \\
 \therefore I_1 &= I_3 = \dots = I_{2023} = \pi.
 \end{aligned}$$

28

sol 1)

$$0 \leq x \leq \frac{\pi}{3} : \sin x < \sqrt{3} \cos x$$

$$\frac{\pi}{3} \leq x \leq \pi : \sin x > \sqrt{3} \cos x.$$

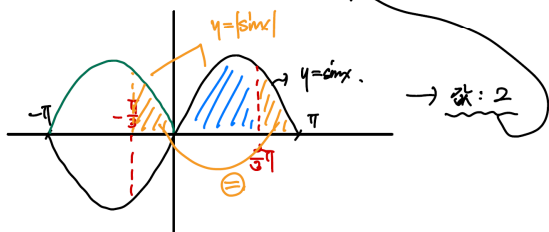
$$\begin{aligned}
 \Rightarrow \int_0^{\frac{\pi}{3}} (-\sin x + \sqrt{3} \cos x) dx + \int_{\frac{\pi}{3}}^{\pi} (\sin x - \sqrt{3} \cos x) dx \\
 = \left[\cos x + \sqrt{3} \sin x \right]_0^{\frac{\pi}{3}} + \left[-\cos x - \sqrt{3} \sin x \right]_{\frac{\pi}{3}}^{\pi} = 4.
 \end{aligned}$$

sol 2)

$$\sin x - \sqrt{3} \cos x \stackrel{\text{정규화}}{=} 2 \sin\left(x - \frac{\pi}{3}\right). \quad \therefore \int_0^{\pi} \left| 2 \sin\left(x - \frac{\pi}{3}\right) \right| dx.$$

$$x - \frac{\pi}{3} = t \Rightarrow dx = dt.$$

$$\int_{-\frac{\pi}{3}}^{\frac{2\pi}{3}} |2 \sin t| dt = 2 \int_{-\frac{\pi}{3}}^{\frac{2\pi}{3}} |\sin t| dt = 4.$$



29

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{2}} (\cos 2x - \sin x)^2 dx \\
 &= \pi \int_0^{\frac{\pi}{2}} (\cos^2 2x - 2\sin x \cos 2x + \sin^2 x) dx \\
 &= \pi \int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 4x}{2} - 2\sin x \cos 2x + \frac{1-\cos 2x}{2} \right) dx
 \end{aligned}$$

$$* \sin(x+2x) = \sin x \cos 2x + \cos x \sin 2x$$

$$+ \sin(x-2x) = \sin x \cos 2x - \cos x \sin 2x$$

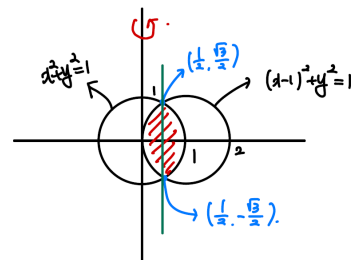
$$\sin 3x - \sin x = 2 \cos x \cos 2x$$

$$\Rightarrow \pi \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} (1 + \cos 4x) - \sin 3x + \sin x + \frac{1}{2} (1 - \cos 2x) \right) dx$$

$$= \pi \left(\frac{\pi}{2} + \frac{2}{3} - \frac{9}{16} \sqrt{2} \right) \quad \therefore a = \frac{1}{2}, b = -\frac{9}{16}, c = \frac{1}{3}$$

$$\therefore 96(a+b+c) = 26$$

30



$$\begin{aligned}
 \therefore V &= 2\pi \left\{ \int_0^{\frac{1}{2}} \sqrt{1-x^2} dx + \int_{\frac{1}{2}}^1 \sqrt{1-x^2} dx \right\} \times 2 = \frac{2}{3}\pi^2 - \frac{\sqrt{3}}{2}\pi \\
 &\quad \downarrow \quad \downarrow \\
 &\quad x = \sin \theta \Rightarrow dx = \cos \theta d\theta \quad \left[-\frac{1}{3}(1-x^2)^{\frac{3}{2}} \right]_{\frac{1}{2}}^1 = \frac{\sqrt{3}}{8} \\
 &\quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos^3 \theta + \cos \theta \sin^2 \theta) d\theta \\
 &= \frac{\pi}{2} - \frac{\sqrt{3}}{4}
 \end{aligned}$$

31

$$\int_0^1 \frac{\cosh x \cdot \sinh^2 x}{1 - \cosh^2 x} dx = \int_0^1 \frac{\cosh x}{\sinh^2 x} dx - \int_0^1 \frac{\cosh x}{\sinh^2 x} \cdot \cosh^2 x dx.$$

$$= \left[\ln |\sinh x| \right]_0^1 - \left[\frac{1}{2} \sinh^2 x \right]_0^1 = \ln(\cosh 1) - \frac{1}{2} \sinh^2 1$$

32

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \left[\frac{1}{n} \sin\left(\frac{\pi k}{n} + \frac{\sqrt{n}}{n}\right) + \frac{1}{n} \sum_{k=1}^n \frac{k}{n} \sin\left(\frac{\pi k}{n} + \frac{\sqrt{n}}{n}\right) \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n \sin\left(\frac{\pi k}{n} + \frac{\sqrt{n}}{n}\right) + \frac{1}{n^2} \sum_{k=1}^n k \sin\left(\frac{\pi k}{n} + \frac{\sqrt{n}}{n}\right) \right] \\ &= \int_0^1 \sin(\pi x) dx = \left[-\frac{1}{\pi} \cos(\pi x) + \frac{1}{\pi^2} \sin(\pi x) \right]_0^1 = \frac{1}{\pi} \end{aligned}$$

33

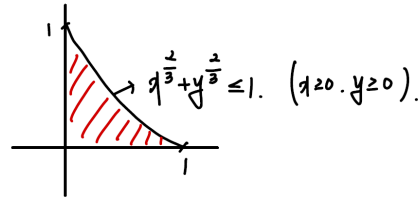
(가) $[0, \frac{\pi}{2}]$ 에서 $\sin x \geq \frac{2}{3}x \Rightarrow \frac{1}{\sin x} \leq \frac{3}{2x}$
 $\therefore \int_0^{\frac{\pi}{2}} \frac{1}{\sin x} dx = \int_0^{\frac{\pi}{2}} \frac{1}{\sin x} dx \geq \int_0^{\frac{\pi}{2}} \frac{2}{3x} dx = \frac{2}{3} \int_0^{\frac{\pi}{2}} \frac{1}{x} dx$
 수렴 (C.T.)

(나) $\frac{\pi}{2} - x = t \Rightarrow -dx = dt$
 $\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin^3 t}} dt = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin^3 t}} dt - \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin^3 t}} dt$
 $\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{1}{t^{\frac{3}{2}}} dt$ 수렴 (C.T.)

$\therefore$ 수렴!
 (다) $\sqrt{dx} \geq t \Rightarrow \frac{1}{\sqrt{dx}} \geq \frac{1}{t} \Rightarrow dx \geq t^2 dt$
 $\int_{\frac{1}{\sqrt{2023}}}^{\infty} \frac{1}{\sqrt{dx}} dt = \int_{\frac{1}{\sqrt{2023}}}^{\infty} \frac{1}{t^2} dt > \int_{\frac{1}{\sqrt{2023}}}^{\infty} \frac{1}{t^2} dt$
 수렴 (C.T.)

(라) $\left[\frac{\sin x}{x} \right]_{2023}^{\infty} + \int_{2023}^{\infty} \frac{\sin x}{x^2} dx$
 $\left| \frac{\sin x}{x^2} \right| \leq \frac{1}{x^2}$
 $\therefore \int_{2023}^{\infty} \frac{\sin x}{x^2} dx$ 수렴 (절대)
 $\therefore$ 수렴 (수렴 + 수렴)

34



Sol I)
 $x = \cos^3 t, y = \sin^3 t \quad (0 \leq t \leq \frac{\pi}{2})$
 $-\int_0^{\frac{\pi}{2}} \sin^3 t \cdot (-3\cos^2 t \sin t) dt = +3 \int_0^{\frac{\pi}{2}} \sin^4 t \cos^2 t dt$
 $= 3 \cdot \left(\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} - \frac{5}{8} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) = \frac{9}{16} \pi - \frac{15}{32} \pi = \frac{3}{32} \pi$

Sol II)
 Astroid $\begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases} \quad (0 \leq t \leq 2\pi)$ 넓이: $\frac{3}{8} \pi a^3$
 $\therefore \frac{3}{8} \pi x \cdot \frac{1}{4} = \frac{3}{32} \pi$

35

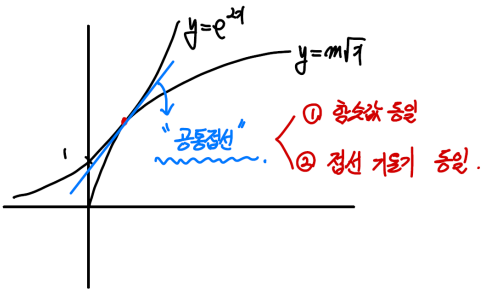
$y = \frac{2}{(x-1)(x+1)(x+1)}$

 $\Rightarrow \int_0^{\frac{1}{2}} \frac{2}{(x-1)(x+1)(x+1)} dx$
 $= 4\pi \cdot \left[-\frac{1}{4} \ln|x-1| + \frac{1}{4} \ln|x+1| - \frac{1}{2(x+1)} \right]_0^{\frac{1}{2}} = (2 - \ln 3) \pi$

36

$\lim_{t \rightarrow \infty} \frac{1}{t^2+t} = 0$
 $\begin{cases} \sin t = \frac{2t}{1+t^2} \\ \cos t = \frac{1-t^2}{1+t^2} \\ dx = \frac{2}{1+t^2} dt \end{cases} \quad (\text{by Weierstrass substitution})$
 $\Rightarrow \int_{\frac{1}{\sqrt{3}}}^1 \frac{1}{t^2+t} dt = \left[\ln t - \ln(1+t) \right]_{\frac{1}{\sqrt{3}}}^1 = \ln \left(\frac{1+\sqrt{3}}{2} \right)$
 $\frac{1}{t(t+1)} = \frac{1}{t} - \frac{1}{t+1}$

37



i) $e^{2a} = m\sqrt{a}$

ii) $2e^{2a} = m \cdot \frac{1}{2\sqrt{a}} = \frac{m}{2\sqrt{a}}$ $\rightarrow 4a=1 \quad a=\frac{1}{4}$

$\Rightarrow e^{\frac{1}{2}} = \frac{m}{2} \Rightarrow m = 2e$

$\therefore \int_0^{\frac{1}{4}} (e^{2x} - 2e \cdot \sqrt{x}) dx = \left[\frac{1}{2}e^{2x} - \frac{4}{3}e \cdot x^{\frac{3}{2}} \right]_0^{\frac{1}{4}}$
 $= \left(\frac{1}{2}e^{\frac{1}{2}} - \frac{4}{3}e \cdot \left(\frac{1}{8}\right) - \left(\frac{1}{2}\right) \right) = \frac{1}{3}e^{\frac{1}{2}} - \frac{1}{3}$

38



$\therefore V = \pi \int_{-10}^{-10+h} x^2 dx \Rightarrow \frac{dV}{dt} = \pi (100 - (-10+h)^2) \cdot \frac{dh}{dt}$

$\frac{dV}{dt} = 3, h=5 \Rightarrow \frac{dh}{dt} = u \Rightarrow 3 = \pi \cdot 75 \cdot u \quad \therefore u = \frac{1}{25\pi} \quad \therefore \eta = 25$

39

$y = \sqrt{f^2(x)+5}$ 의 역함수를 구해보자.

$y = \sqrt{f^2(x)+5} \Rightarrow x^2 = f^2(y)+5 \Rightarrow f^2(y) = x^2-5$

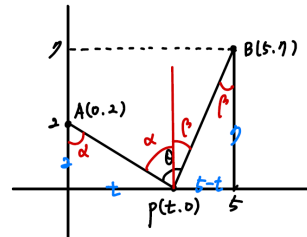
$\therefore f(y) = \sqrt{x^2-5} \Rightarrow y = \underset{\substack{\uparrow \\ \text{f}}}{f^{-1}(\sqrt{x^2-5})}$

$\sim \sqrt{f^2(x)+5}$ 과 $g(\sqrt{x^2-5})$ 는 역함수 관계

$\therefore \int_3^5 g(\sqrt{x^2-5}) dx + \int_{\sqrt{f^2(2)+5}}^{\sqrt{f^2(5)+5}} \sqrt{f^2(x)+5} dx = 10.$
 $\Delta = g(2) = f^{-1}(2)$
 $\Delta = g(2) = f^{-1}(2)$

$\therefore \int_3^5 g(\sqrt{x^2-5}) dx = 3$

40



$\therefore \tan \theta = \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = \frac{\frac{t}{2} + \frac{5-t}{1}}{1 - \frac{t}{2} \cdot \frac{5-t}{1}} = \frac{5t+10}{t^2-5t+14} \triangleq f(t).$

$\Rightarrow f'(t) = \frac{(-5) \cdot (t^2+4t-24)}{(t^2-5t+14)^2} = 0 \quad \therefore t = -2+2\sqrt{7} (t>0).$

$t=0$ & $t=5$ 일때도 check! ... 생략!